INVARIANCE AT COINCIDING POINTS

Abstract. A bounded multilinear form on vector-valued $L^2(\mathbb{R}^3)$ vanishes if it vanishes on entries with disjoint planar supports and whenever one entry is a compactly supported gradient. We prove this statement and apply it to moments of closed, Gaussian-dominated two-form fields. It extends rotation identities from separated planar supports to all L^2 tests and hence to the law of the field.

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1. Introduction

Write $x = (y, z) \in \mathbb{R}^2 \times \mathbb{R}$, and, for an open set $U \subset \mathbb{R}^2$, put

$$\mathcal{L}_U = \{f \in L^2(\mathbb{R}^3; \mathbb{C}^3) : f = 0 \text{ almost everywhere outside } U \times \mathbb{R}\}.$$

All norms on these spaces are L^2 norms. We say that a field lies over $K \subset \mathbb{R}^2$ when it vanishes almost everywhere outside $K \times \mathbb{R}$. Separation therefore concerns the projected planar supports; the transverse coordinate z remains smeared.

Theorem A (Absence of contact terms)**.** Let $U \subset \mathbb{R}^2$ be open and $N \geq 1$. Suppose that $D\colon \mathcal{L}_U^N \to \mathbb{C}$ is a complex multilinear form satisfying

$$|D(f_1, \dots, f_N)| \leq M \prod_{a=1}^{N} \|f_a\|_2 \qquad (f_a \in \mathcal{L}_U)$$

for some $M < \infty$. Assume that:

(i) $D(f_1, \dots, f_N) = 0$ if the entries lie over compact sets $K_1, \dots, K_N \subset U$ that are pairwise disjoint;

(ii) $D(f_1, \dots, f_N) = 0$ whenever one entry is ∇u for some $u \in C_c^\infty(U \times \mathbb{R}; \mathbb{C})$.

Then $D = 0$. ■□

An identity for correlation functions at separated arguments need not determine the same identity at coinciding arguments. In a Gaussian-dominated field theory the moments are bounded multilinear forms on L^2, so their values are defined even when the test supports overlap. The gradient hypothesis above is the additional condition supplied by closedness of a two-form in three dimensions.

The theorem is Proposition 8.4 of [3]. No probability law enters its hypotheses. Its application to rotations is direct: the difference between a rotated and an unrotated moment is bounded, and closedness makes it vanish on gradients. The remaining hypothesis is precisely the invariance already obtained away from coinciding planar arguments.

The proof has two different steps. For a bilinear form, locality forces its representing operator to commute with multiplication by functions of y. One can then multiply gradients by arbitrary

bounded planar functions and obtain a total family of vector fields. For three or more entries, induction removes terms involving disjoint groups of supports. A partition into small squares leaves only terms for which all entries lie over the same square; their sum tends to zero by the L^2 bound. Both steps are needed. Boundedness alone does not remove bilinear forms supported on the diagonal.

2. Locality and gradients

We prove theorem A simultaneously for all open sets U, by induction on N. A bounded Borel function h on $\mathbb{R}^2$ acts on $\mathcal{L}_U$ by

$$(hf)(y,z) = h(y)f(y,z).$$

For a Borel set A, the same notation 1_A denotes its indicator and the corresponding orthogonal projection. Fields lying over compact subsets of U are dense in $\mathcal{L}_U$: take an increasing compact exhaustion of U and use dominated convergence. When $N = 1$, the first hypothesis of the theorem therefore gives the conclusion.

For $N = 2$, apply the Hilbert-space representation theorem [2, Theorem 3.3] to the bounded sesquilinear form $(f', f) \mapsto D(\overline{f'}, f)$. There is a bounded operator T on $\mathcal{L}_U$ such that

$$D(f_1, f_2) = \int_{U\times\mathbb{R}} f_1(x) \cdot (Tf_2)(x)\,\mathrm{d}x, \tag{1}$$

where the dot denotes the complex bilinear pairing on $\mathbb{C}^3$.

The first hypothesis implies $1_K T 1_{K'} = 0$ for disjoint compact subsets K, K' of U. Indeed, its matrix elements in (1) are $D(1_K f_1, 1_{K'} f_2)$, which vanish. The same conclusion holds for disjoint Borel subsets A, A' of U. To see this, use inner regularity of planar Lebesgue measure on the bounded sets $A \cap \{|y| < n\}$ and $A' \cap \{|y| < n\}$ to choose increasing compact sets $K_n \subset A$ and $K_n' \subset A'$ whose unions exhaust A and A' up to null sets. Then $1_{K_n} \to 1_A$ and $1_{K_n'} \to 1_{A'}$ strongly on $\mathcal{L}_U$, so

$$1_A T 1_{A'} = \lim_{n\to\infty} 1_{K_n} T 1_{K_n'} = 0$$

in the strong operator topology.

Take $A' = U \setminus A$ and then interchange the two sets. The resulting identities give

$$T1_A = 1_A T 1_A = 1_A T.$$

Since every bounded Borel function is a uniform limit of simple Borel functions, T commutes with all planar multipliers. Consequently

$$D(hf_1, f_2) = D(f_1, hf_2). \tag{2}$$

The gradient hypothesis and this identity imply

$$D(h\nabla u, f) = 0, \qquad u \in C_c^\infty(U \times \mathbb{R}; \mathbb{C}), \quad f \in \mathcal{L}_U, \tag{3}$$

for every bounded Borel function h on $\mathbb{R}^2$.

We now use (3) to separate the three components of a vector field. Fix a compact set $A \subset U$, choose $u_0 \in C_c^\infty(U)$ equal to one near A, and let $\chi \in C_c^\infty(\mathbb{R})$. Write $y = (y^0, y^1)$. The gradients of

$$u_0(y)\chi(z), \qquad u_0(y)y^0\chi(z), \qquad u_0(y)y^1\chi(z)$$

restrict on $A \times \mathbb{R}$ to

$$\chi'(z)e_3, \qquad \chi(z)e_1 + y^0\chi'(z)e_3, \qquad \chi(z)e_2 + y^1\chi'(z)e_3,$$

respectively. Apply (3) with the multiplier $h1_A$. The first gradient gives

$$D(h1_A\chi' e_3, f) = 0.$$

In the second identity the term containing e_3 vanishes by this formula, with the bounded multiplier $hy^0 1_A$. The third identity is treated with $hy^1 1_A$. Thus

$$D(h1_A\chi e_1, f) = D(h1_A\chi e_2, f) = D(h1_A\chi' e_3, f) = 0. \tag{4}$$

The fields in (4), as A, h, χ vary, have dense linear span in $\mathcal{L}_U$. For the first two components this is the density of products of planar functions of compact support and smooth compactly supported functions of z. For the third component, the derivatives χ' are themselves total in $L^2(\mathbb{R})$. A function orthogonal to every such derivative has zero distributional derivative, and hence is almost everywhere constant [1, Proposition 1.4.8]; the only constant in $L^2(\mathbb{R})$ is zero. It follows that the continuous form $D(\cdot, f)$ vanishes on all of $\mathcal{L}_U$. This proves the theorem for two entries.

3. Several coinciding arguments

Let $N \geq 3$, and suppose that the theorem holds for every smaller number of entries and every open subset of $\mathbb{R}^2$. Before partitioning the supports, we record two consequences of the induction hypothesis.

First, fix an index i and a compact set $K \subset U$. If f_i lies over K and all the other entries belong to $\mathcal{L}_{U\setminus K}$, then

$$D(f_1, \dots, f_N) = 0. \tag{5}$$

Indeed, hold f_i fixed. The remaining entries define a bounded $(N-1)$-linear form on $\mathcal{L}_{U\setminus K}$. It vanishes on gradients there, because a compactly supported smooth function on $(U \setminus K) \times \mathbb{R}$, extended by zero, belongs to $C_c^\infty(U \times \mathbb{R}; \mathbb{C})$. It also vanishes on entries over pairwise disjoint compact subsets of $U \setminus K$: those compacts, together with K, satisfy the first hypothesis for D. Induction gives (5).

Second, partition the indices into two nonempty sets I, J, and let $K_J \subset U$ be compact. If f_j lies over K_J for every $j \in J$ and $f_i \in \mathcal{L}_{U\setminus K_J}$ for every $i \in I$, then

$$D(f_1, \dots, f_N) = 0. \tag{6}$$

Hold the entries in J fixed and regard D as a form with $|I| < N$ entries on $\mathcal{L}_{U\setminus K_J}$. The gradient hypothesis is inherited. To check the other hypothesis, suppose that the entries with indices in I lie over pairwise disjoint compact sets K_i in $U \setminus K_J$. Choose one $i' \in I$. Every other entry lies outside $K_{i'} \times \mathbb{R}$, so (5), with $K = K_{i'}$, gives vanishing. Induction applied to this $|I|$-linear form proves (6).

By density and continuity, it remains to treat entries that all lie over one compact set $K_0 \subset U$. For $d > 0$, let

$$\mathcal{Q}_d = \{d(k + [0,1)^2) : k \in \mathbb{Z}^2\}.$$

Only finitely many of these half-open squares meet K_0. Multilinearity therefore gives

$$D(f_1, \dots, f_N) = \sum_{Q_1, \dots, Q_N \in \mathcal{Q}_d} D(1_{Q_1} f_1, \dots, 1_{Q_N} f_N). \tag{7}$$

The sum has only finitely many nonzero terms. We claim that every term with squares Q_a not all equal vanishes.

For such a tuple put $I = \{i : Q_i = Q_1\}$ and let J be its nonempty complement. For $0 < \varepsilon < d/2$, denote by Q^ε the closed square of side $d - 2\varepsilon$ concentric with Q. The compact sets

$$K_0 \cap Q_1^\varepsilon, \qquad K_J = K_0 \cap \bigcup_{j \in J} Q_j^\varepsilon$$

are disjoint. The entries $1_{Q_j^\varepsilon} f_j$, $j \in J$, lie over K_J, and the entries with indices in I lie outside K_J. Equation (6) gives

$$D(1_{Q_1^\varepsilon} f_1, \dots, 1_{Q_N^\varepsilon} f_N) = 0.$$

For each a, $1_{Q_a^\varepsilon} f_a \to 1_{Q_a} f_a$ in L^2 as $\varepsilon \downarrow 0$, since the boundary of a square has measure zero. Continuity proves the claim.

Only the terms in which all entries lie over the same square remain in (7). Their sum satisfies

$$|D(f_1,\dots,f_N)| \le M \sum_{Q\in\mathcal{Q}_d} \prod_{a=1}^{N} \|1_Q f_a\|_2$$
$$\le M \left(\max_{Q\in\mathcal{Q}_d} \|1_Q f_3\|_2\right) \prod_{a\neq 3} \|f_a\|_2. \tag{8}$$

For the last inequality use Cauchy–Schwarz in the sum for the first two entries and $\|1_Q f_a\|_2 \le \|f_a\|_2$ for $a \ge 4$. The function

$$y \longmapsto \int_{\mathbb{R}} |f_3(y,z)|^2 \,\mathrm{d}z$$

belongs to $L^1(\mathbb{R}^2)$ by Tonelli's theorem. Each square has area d^2, so absolute continuity of its integral gives

$$\max_{Q\in\mathcal{Q}_d} \|1_Q f_3\|_2 \longrightarrow 0 \qquad (d \downarrow 0).$$

The left side of (8) is independent of d. It is therefore zero. This completes the induction and the proof of theorem A. □

4. The rotation defect

For $g \in O(3)$ we use the action on pseudovector fields

$$(g\cdot b)(x) = \det(g)\, g\, b(g^{-1}x). \tag{9}$$

This is the action obtained from two-forms under the Hodge isometry $*f = (f_{23}, -f_{13}, f_{12})$. Let R_t be rotation by t in the (x_1,x_2)-plane, fixing x_3.

Corollary 4.1 (From separated supports to all tests)**.** Let $N \ge 1$, and let S_N be a bounded complex N-linear form on $L^2(\mathbb{R}^3;\mathbb{C}^3)$. Assume that it vanishes whenever one entry is ∇u, $u \in C_c^\infty(\mathbb{R}^3;\mathbb{C})$. Fix $t \in \mathbb{R}$, and suppose that

$$S_N(R_t\cdot b_1,\dots,R_t\cdot b_N) = S_N(b_1,\dots,b_N)$$

whenever the entries lie over pairwise disjoint compact subsets of $\mathbb{R}^2$. Then the same identity holds for all entries in $L^2(\mathbb{R}^3;\mathbb{C}^3)$. ■□

Proof. Apply theorem A, with $U = \mathbb{R}^2$, to

$$D_t(b_1,\dots,b_N) = S_N(R_t\cdot b_1,\dots,R_t\cdot b_N) - S_N(b_1,\dots,b_N).$$

The action of R_t is an L^2 isometry, so D_t is bounded. The first vanishing hypothesis is the assumed separated-support identity. For the second, use

$$R_t\cdot\nabla u = \nabla(u\circ R_{-t}).$$

Since $u\circ R_{-t} \in C_c^\infty(\mathbb{R}^3;\mathbb{C})$, both terms defining D_t vanish when an entry is ∇u. □

For the application to curvature, let μ be a probability law of real tempered two-forms F on $\mathbb{R}^3$, with the sigma-algebra generated by evaluations on real Schwartz tests, and put $\mathbb{B}(b) = F(*^{-1}b)$ on Schwartz vector fields. Gaussian domination means

$$\int e^{\mathbb{B}(b)}\,\mathrm{d}\mu \le e^{\|b\|_2^2/2} \qquad (b \in \mathcal{S}(\mathbb{R}^3;\mathbb{R}^3)). \tag{10}$$

Closedness means $dF = 0$ almost surely. In vector notation it gives

$$\mathbb{B}(\nabla u) = 0 \qquad (u \in C_c^\infty(\mathbb{R}^3;\mathbb{C})) \tag{11}$$

almost surely; see [3, Section 8.5] for this vector form of closedness.

By [3, Proposition 5.1], the map $b \mapsto \mathbb{B}(b)$ extends to a continuous complex linear map from L^2 to every $L^p(\mu)$, $1 \le p < \infty$. Its moments

$$S_N(b_1, \dots, b_N) = \int \mathbb{B}(b_1) \cdots \mathbb{B}(b_N) \, \mathrm{d}\mu$$

are bounded symmetric multilinear forms. More precisely,

$$\|\mathbb{B}(b)\|_{L^N(\mu)} \le c_N \|b\|_2, \qquad |S_N(b_1, \dots, b_N)| \le c_N^N \prod_a \|b_a\|_2,$$

with constants c_N depending only on N. The moment bound follows from the field bound by Hölder's inequality. In particular no condition on the relative supports is needed to define S_N. Equation (11) gives the gradient hypothesis of corollary 4.1.

Proposition 4.2 (Invariance of a closed Gaussian-dominated law)**.** Suppose that μ satisfies (10) and is almost surely closed. Assume that, for every $N \ge 1$ and every $t \in \mathbb{R}$, its moments satisfy the rotation identity of corollary 4.1 on entries over pairwise disjoint compact planar sets. Then all moments satisfy that identity on $L^2(\mathbb{R}^3; \mathbb{C}^3)$, and $(R_t)_* \mu = \mu$ for every t. ■

Proof. The moment bounds and (11) verify the hypotheses of corollary 4.1, which proves the first assertion. To recover the law, take a real Schwartz field b. The partial sums of the exponential series for $e^{i\mathbb{B}(b)}$ have modulus at most

$$e^{|\mathbb{B}(b)|} \le e^{\mathbb{B}(b)} + e^{-\mathbb{B}(b)}.$$

The last expression is integrable by (10). Dominated convergence therefore gives

$$\int e^{i\mathbb{B}(b)} \, \mathrm{d}\mu = 1 + \sum_{N \ge 1} \frac{i^N}{N!} S_N(b, \dots, b).$$

The series for $R_t \cdot b$ has the same terms by the first assertion. The characteristic functional is thus invariant. Laws on tempered distributions are determined by their characteristic functionals, so $(R_t)_* \mu = \mu$; see also [3, Section 5.2]. □

This is the final step in the proof of [3, Theorem 8.6]. For that application, angular covariance of the plate kernels and their identification as moment densities on all distinct configurations [3, Sections 8.9–8.10] give the separated-support identity. Theorem A then uses the bounded moments and their gradient identities to remove the support restriction.

5. What the domain specifies at coincidence

The term "invariance at coinciding arguments" must be read through the domain of the correlation functional. For a distribution defined on the open configuration space

$$(\mathbb{R}^d)^N_{\neq} = \{(x_1, \dots, x_N) : x_a \neq x_b \text{ for } a \neq b\},$$

covariance is an identity on that open set. It gives no values of an extension on the diagonals. A formulation on Schwartz functions flat at every diagonal likewise imposes its identities on that stated test-function space. A functional given only on separated compactly supported C^1 tests likewise specifies no values on overlapping supports. This domain should be kept distinct from the unrestricted L^2 moments used here.

In the Gaussian-dominated three-dimensional setting, overlapping supports are already allowed before invariance is proved. It remains to extend the covariance identity from tests with disjoint planar supports to all L^2 tests: the analytic plate kernels are first identified at pairwise distinct planar points. The corresponding rotation defect may still be supported where some planar arguments coincide. Theorem A excludes that defect by combining the L^2 bound with the gradient identities. It does not assign a pointwise value to a singular correlation kernel on the diagonal. Its conclusion is an identity between bounded multilinear forms for all L^2 test fields.

References

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